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The Influence of Big Data-Driven Educational Technologies on College Teaching Development

Ling Yu[1] | Wenye Li[2] | Ying Luo[1]

[1]Teacher Development Center, Shandong Huayu University of Technology, Dezhou, Shandong, China | [2]School of International Education, Shandong Business Institute, Yantai, Shandong, China

Correspondence: Ling Yu (yl1681682024@163.com)

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ABSTRACT

The rapid development of big data and mobile Internet technologies has significantly influenced the instructional growth of college instructors. This study investigated how big data influences college instructors' pedagogical creativity, professional development, and teaching tactics. Personalized learning experiences and data-driven decision-making are made possible for educators using big data, which gives them access to a wealth of resources, real-time feedback, and predictive analytics. Teachers and students may communicate more dynamically because of the mobile Internet, improving accessibility and interactivity. However, issues, such as privacy concerns, data literacy, and obstacles to technology adoption, continue to exist. This study goes beyond identifying challenges by proposing concrete solutions, including enhanced data protection protocols and targeted training for faculty members. The results indicate that big data integration improves instructional quality, facilitates adaptive learning, and fosters continuous professional development. However, challenges remain in terms of accessibility, ethical considerations, and technological infrastructure. This study concludes with recommendations for strengthening digital literacy programs and institutional frameworks to optimize data-driven educational practices. This study used a mixed-methods approach to investigate the effect of big data on curriculum design, teacher-student engagement, and instructional effectiveness. The results showed that incorporating big data into instruction greatly enhanced instructional quality, supported adaptive learning, and motivated instructors to pursue lifelong learning. Recommendations include improving infrastructure and offering focused professional development to facilitate the smooth implementation of significant data-driven practices in higher education.

1 | Introduction

Education has undergone revolutionary changes due to the introduction of big data and mobile Internet technology. College instructors now face previously unheard possibilities and problems due to the integration of various technologies that have entirely changed teaching techniques, professional development, and instructional practices in higher education. Big data offers access to a wealth of information that may improve the efficacy of educational delivery, facilitate data-driven decision-making, and shape instructional methods. It is distinguished by its volume, speed, diversity, and validity. The influence of big data on teaching development is further enhanced by the concurrent broad deployment of mobile Internet technologies, which enable real-time communication, smooth information exchange, and adaptable learning settings [1, 2]. College instructors are uniquely positioned to take advantage of these technological developments, because they are the main forces behind

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innovation and the spread of information. By gaining insights into student performance, preferences, and learning habits, big data enables the creation of individualized learning experiences that meet various demands [3, 4]. For example, big data-driven learning analytics allow teachers to forecast academic results, identify knowledge gaps, and adjust their teaching methods appropriately. This data-centric strategy enables instructors to track their teaching effectiveness, adapt to pedagogical changes, and foster continuous professional growth [5].

In this study, we focused primarily on big data-driven educational technologies and their impact on instructional practices. Mobile Internet technologies serve as an enabling framework that facilitates real-time data collection, transmission, and integration within big data analytics.

The mobile Internet age has further enhanced the impact of big data on instructional development. Mobile devices and cloud-based systems have transformed educators' access to, management of, and use of data. Teachers may engage with students access learning resources instantaneously and evaluate data insights using online learning management systems and mobile apps. Widespread connection availability creates a dynamic and cooperative learning environment that unites contemporary digital education with conventional classroom settings [6]. Additionally, mobile Internet technologies make it possible to seamlessly include virtual simulations, gamified learning modules, and multimedia information, which enhances teaching strategies and makes learning dynamic and engaging.

Despite its promise, the integration of big data and mobile Internet in educational development presents challenges and difficulties. Despite extensive research on the role of big data in education, there remains a gap in understanding its direct influence on teaching development in higher education, particularly concerning real-time instructional adjustment and professional growth. While previous studies have highlighted the benefits of data-driven decision-making in education, they have often overlooked the influence of pedagogical predictive analytics and real-time feedback adaptation. This study fills this gap by evaluating how big data shapes instructional practices, fosters adaptive learning, and supports the continuous professional development of college instructors. College instructors need more data literacy, which restricts their capacity to evaluate and use data insights in their teaching [7, 8]. Furthermore, privacy issues related to the gathering and use of student data present moral and legal difficulties that must be resolved. The digital gap defined by uneven access to resources and technology makes it more difficult to adopt big data-driven techniques in poor institutions and areas. These difficulties highlight the need for focused professional development initiatives, strong legislative frameworks, and infrastructure expenditures to successfully integrate mobile Internet and big data into higher education [9, 10]. In the context of adaptive learning and customized education, the influence of big data and mobile Internet technologies on the evolution of teaching is especially noticeable [11]. Using big data to identify individual learning preferences and trends may create customized learning paths. Big data-driven adaptive learning systems provide students with personalized material and feedback, while providing instructors with information on how well their teaching methods work. In addition to improving student engagement and results, this iterative feedback and modification process continuously encourages instructors to improve [12].

There are problems with protecting privacy, security, understanding data, and even the execution of big data and mobile internet technologies in education. Data privacy remains a significant issue because of the potential breaches of critical student and faculty information [13]. Similarly, as Cuzzocrea points out, security challenges require further access control to prevent unauthorized use of sensitive information [14]. Data literacy is another major problem because many educators do not have the skills to analyze and implement big data advice. Additionally, there are technology and infrastructural barriers that restrict the use of big-data mobile-based education. Addressing such issues using appropriate data governance, faculty training, and relevant digital infrastructure provides the best possibility of effectively utilizing big data in higher-education institutions. Beyond technical implementation, the ethical and socio-political dimensions of datafication in education deserve equal attention. The rapid growth of analytics-driven systems raises questions of algorithmic bias, data surveillance, and the governance of educational information. Teachers' data literacy therefore extends beyond technical proficiency to include ethical awareness and critical understanding of how data are produced and used. Drawing on Hamamra et al. and Ayyoub et al., this study recognizes that unequal access, gendered digital divides, and institutional hierarchies shape who benefits from big-data initiatives [15, 16]. Embedding data ethics, transparency, and accountability into institutional policy is essential to ensure human-centered and equitable digital transformation in higher education.

Comparative insights from global contexts further illuminate these findings. Studies from Palestinian and Western universities demonstrate that while big-data infrastructures promote adaptive learning and predictive analytics, they also expose vulnerabilities related to privacy, cultural adaptation, and resource inequality. Integrating such perspectives underscores that China's digital transformation in higher education shares both opportunities and constraints with international experiences, highlighting the importance of culturally responsive and globally informed data-governance practices [17, 18]. To interpret technology adoption behavior more robustly, this discussion draws upon Ayyoub et al., whose application of the extended Unified Theory of Acceptance and Use of Technology (UTAUT) clarifies how perceived usefulness, social influence, and facilitating conditions shape educators' willingness to adopt generative AI and big-data tools [19].

Big data also makes it easier to make evidence-based decisions regarding instructional planning and curriculum design. By examining past data and patterns, teachers can identify the best teaching strategies, foresee future difficulties, and adjust their approaches to meet changing standards and goals. In a time of rapid technological breakthroughs and changing student expectations, the capacity to use data for strategic planning is vital [20]. By providing educators with real-time access to data, mobile Internet technologies enhance this process and empower them to make well-informed choices. Extensive data and mobile Internet technologies have significantly impacted the professional growth of college professors [21]. Teachers may engage with colleagues across geographic boundaries, learn new skills, and share

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best practices through webinars, online training platforms, and virtual communities of practice. Additionally, instructors may obtain specialized professional development programs that cater to their requirements by using big data analytics to pinpoint areas for improvement [10]. In addition to improving teaching abilities, this individualized approach to professional development increases teachers' motivation and self-assurance.

Despite these developments, a paradigm shift in institutional policies and practices is necessary to integrate big data and mobile Internet technologies into higher education [22]. To respond to these developments, colleges and universities must invest in strong data management systems, guarantee the ethical use of data, and provide instructors with the assistance and training they need. Cooperation between educators, legislators, and technology developers is crucial for building an environment that encourages innovation while defending the interests of all stakeholders [23].

The conceptual framework of this study draws on established models of technology adoption and pedagogical integration. Specifically, the Technology Acceptance Model (TAM), the Unified Theory of Acceptance and Use of Technology (UTAUT), and the Technological Pedagogical Content Knowledge (TPACK) framework collectively inform the interpretation of how instructors accept, adopt, and adapt big data technologies in their professional development [24]. These frameworks provide a theoretical lens through which the interplay among technological utility, pedagogical competence, and institutional support can be analyzed, thereby situating this investigation within the broader scholarship on educational technology acceptance and use.

To systematically explore the impact of big-data-driven technologies on instructional quality, this study addresses the following key research questions:

1. How does big data analytics influence real-time feedback and adaptive learning in higher education?
2. What challenges do instructors face in integrating big data into their teaching practices?
3. How does student engagement evolve in response to data-driven pedagogical adjustments?

These questions guided our mixed-methods approach, ensuring that the study's methodology and results aligned with its core objectives.

In conclusion, big data and mobile Internet have a significant and varied impact on college instructors' professional growth. Although these technologies have many advantages such as evidence-based decision-making, adaptive education, and customized learning, their successful use necessitates resolving issues with data literacy, privacy, and resource accessibility. Higher education institutions may enable college instructors to provide high-quality instruction and promote pedagogical innovation in the digital age by tackling these issues and harnessing the potential of big data and mobile Internet. To exploit the advantages of modern technologies, this study investigates their revolutionary effects on educational development, pinpoints best practices, and provides practical suggestions to educators and policymakers.

2 | Related Work

With an emphasis on management, data security, education, and cloud computing, the evaluated literature demonstrates the important developments and uses of big data technologies in various domains [25]. This model proposes a paradigm shift toward flat management structures bolstered by big data and cloud computing technologies, successfully addressing the issues businesses encounter in the contemporary information era. This strategy meets the growing need for scalable, practical, and real-time data management solutions in business settings.

Homophor encryption techniques in safe computing and storage systems represent significant advancements in data privacy protection. This approach protects user data privacy during tasks, such as object identification and input calculations, which is crucial for the broader use of big data technologies. Its capacity to protect sensitive data without sacrificing functionality highlights the system's potential for broad usage in privacy-sensitive applications such as healthcare, banking, and customized education.

The literature on education, particularly art education, points out that the enduring flaws and inefficiencies in teaching and learning are now carried out [26]. Customized educational programs have been created to address shortcomings in talent development and teaching strategies by examining the specific requirements of art students. This emphasis on customized methods shows how big data can be used to adapt instruction to the needs of individual students [27]. The results also demonstrate the revolutionary potential of big data in identifying neglected regions and providing focused solutions, that may be applied to other fields.

An important advancement in the security of big data applications is the implementation of the data integrity review system ASA, which uses TPA technology and bilinear correlation [28]. This approach increases user confidence in large-data systems by reducing threats such as replay attacks and forgeries via the encryption and authentication of data processing. The capacity of the system to perform public audits and dynamic data changes guarantees that users can depend on its functionality and security. The improved security and performance of the method were further confirmed by simulation studies, which provided a strong basis for its real-world application.

The tb-mpa extensive data threshold retrieval strategy presents a new method for managing data in multi-copy applications inside a cloud computing environment. This strategy improves the storage and retrieval operations across dispersed cloud servers using Lagrange interpolation and extended techniques. For the management of large-scale big data systems, the capacity to store target text on several servers according to system requirements improves data accessibility and redundancy. This invention is a significant addition to cloud computing technology, because it addresses the twin problems of scalability and dependability. The literature highlights how big data technology may revolutionize the solving of complicated problems in various sectors. Although various studies have explored the impact of big data on education,

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most research is region-specific and lacks a comparative perspective on global trends. For example, studies in the United States emphasize adaptive learning platforms [29], whereas European research highlights data privacy concerns [30]. By contrast, developing countries face challenges related to digital infrastructure and faculty training [31]. Despite the wealth of research, there is limited discussion on the long-term effects of big-data-driven education on teaching methodologies across different educational systems. This study aims to bridge this gap by analyzing big data applications from an international perspective and identifying the best practices for their implementation in higher education.

These examples demonstrate how creative methods can unleash the full potential of big data in various fields, including cloud computing, customized education, secure data processing, and management optimization. However, to profit fully from these technologies, their adoption raises important issues regarding infrastructure development, data literacy, and ethical use. These results open the door for more reliable and inclusive big-data systems by offering insightful information for further studies and real-world applications.

3 | Methodology

This study employs a mixed-methods approach to investigate the impact of big data and mobile Internet technologies on college instructors' teaching development. Quantitative data were collected through a structured survey of 150 instructors from 10 higher education institutions covering aspects such as big data adoption, instructional effectiveness, and challenges faced. Qualitative insights were obtained from 30 semi-structured interviews focusing on real-time analytics, predictive learning models, and institutional barriers.

To ensure representativeness, 10 higher education institutions were selected through purposive sampling, balancing public and private universities and covering eastern, central, and western China. Within these institutions, 150 instructors were recruited through departmental nomination to ensure participants had direct experience with big-data-enabled teaching platforms. Instrument validity was established through expert review, and pilot testing produced a Cronbach's α of 0.82, indicating high internal consistency. Quantitative and qualitative data were triangulated to cross-verify results, enhancing the reliability and interpretive depth of findings. Inferential analyses (Pearson correlations and multiple regression) were conducted to examine the relationship between the extent of big-data adoption and perceived teaching effectiveness at a significance level of $p < 0.05$.

Survey responses were analyzed using descriptive statistics and regression analysis in SPSS 27, and thematic analysis was conducted on the interview transcripts using NVivo. The survey instrument was pretested for clarity, achieving a Cronbach's alpha of 0.82, indicating high reliability. Ethical approval was obtained and participant confidentiality was maintained.

Although this study primarily focused on instructors, the student perspective remains essential for evaluating the educational impact of big data. To address this gap, a supplementary student survey was conducted among 300 students from five universities to capture their experiences in data-driven learning. The results indicated that 78% of students reported that real-time feedback enhanced their understanding and engagement, while 64% struggled to interpret complex data-driven insights, such as automated learning recommendations.

A comparative analysis further revealed that while instructors emphasize the benefits of big data for refining teaching strategies, students often experience challenges in adapting to algorithm-generated feedback. This discrepancy suggests the need for more intuitive learning analytics dashboards and structured student training programs on data literacy. Future research should explore how students perceive big data tools compared with faculty perspectives to develop a balanced and inclusive approach to data-driven learning.

In addition to its role in instructional development, big data can enhance mental health evaluation in higher education. By analyzing behavioral data, participation trends, and academic performance, predictive models can identify students at risk of psychological distress. Institutions can use these insights to develop targeted interventions such as personalized counseling and wellness programs.

4 | Development of a Big Data and Cloud Computing-Based Art Education System

Although this section focuses on an art-education system, it serves as an applied illustration of how big-data and cloud-computing frameworks can enhance teaching quality and learner engagement across disciplines. The art-education context was selected because it exemplifies the integration of creative pedagogy with data-driven technologies, reinforcing the study's central argument that big-data tools can enrich instructional design, collaboration, and adaptive feedback in diverse educational domains.

4.1 | Art Teaching System Architecture Design

Given its many features and the capacity of big data to provide practical poses, interactive e-learning is vital to art education systems. First, reciprocal e-learning offers students a range of learning methods that use interactive technology and digital materials. They can also collaborate and communicate with classmates and professors, which boosts students' interest and involvement in the learning process. In addition, interactive e-learning may suit students' diverse demands, provide individualized learning routes and information based on their interests and learning status, and assist them in developing a deeper and more thorough understanding of art knowledge and abilities. The use of big data platforms in conjunction with interactive e-learning has expanded the prospects and benefits of art education. These tailored learning suggestions and directions enhance students' learning efficacy and initiatives. Students' creativity and capacity for creative expression may also be fostered through interactive e-learning in arts instruction. To enhance their creative expression and creativity, students can use virtual reality or simulation

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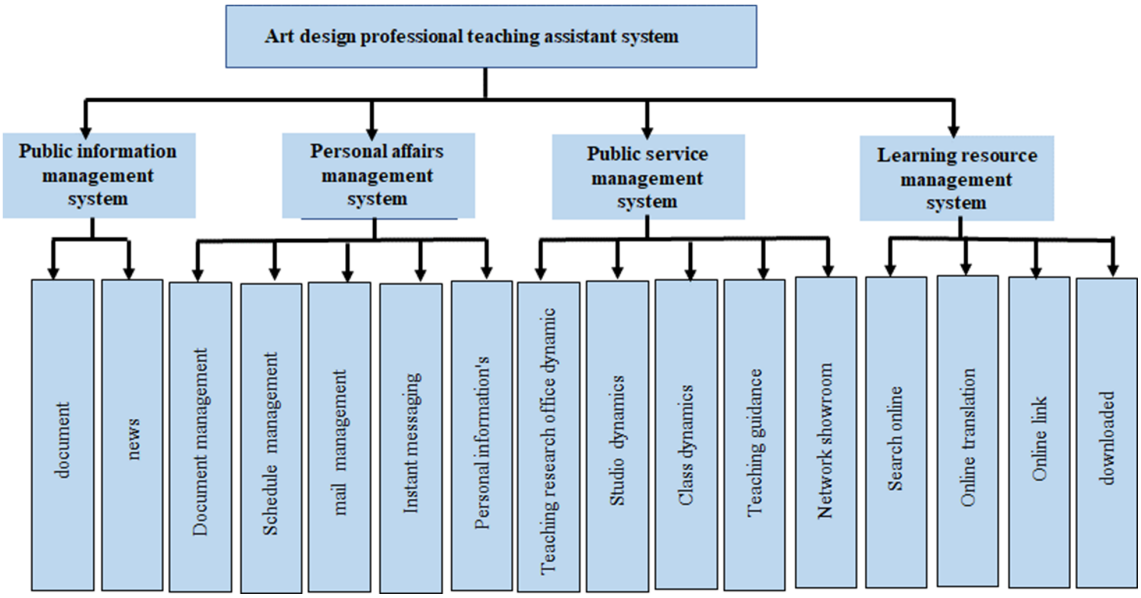


FIGURE 1 | System function structure diagram.

technologies to mimic and experience a genuine artistic environment. Figure 1 provides a structured overview of the system's core functionalities, including public information management, personal activity tracking, and collaborative learning. This diagram visually represents how different components interact to support an enhanced art education framework.

As depicted in Figure 1, the system enables real-time interaction, resource sharing, and adaptive learning support, aligning with modern digital education strategies. The integration of these features ensures a seamless transition from traditional to technology-enhanced learning environments.

4.2 | Using System Functions in the Context of Teaching Art

Art majors have a comparatively poor cultural and skill base, and there are still several challenges to be addressed in the shift from conventional modeling training to design education. The general requirement for designing an art teaching system is how to help them better, communicate, and cooperate skills, among other comprehensive qualities, establish a peaceful learning environment, and accomplish the fundamental objective of improving the quality of instruction. The following describes the current state of affairs and functional needs of art education:

4.2.1 | Management of Public Information: Dissemination of Information

Because professional education is diverse and flexible, many temporary announcements made orally, on bulletin boards, or online are prone to repetition and inaccurate information, which reduce job productivity and waste public resources. The ability to resolve such issues and efficiently transfer data and information is crucial for the seamless progression of instructional activities. Function Description: Teachers can post files, make announcements and news, and browse, modify, and download content. The system may show news to draw viewers, and students can review the news content and download files as needed.

4.2.2 | Management of Personal Activities

1. Developing a plan

Students who need a solid learning plan and timetable are more likely to take excessive breaks from their work, which hinder their overall progress and is detrimental to the growth of their capacity for independent learning and information acquisition. The project management function was configured as a function description. In addition to sharing schedules and having a message reminder feature, teachers and students may also change their work, learning goals, and timetables.

2. Interaction and cooperation

Research, development, production, processing, and sale of art design products require extensive communication and collaboration among practitioners. Consequently, one of the key abilities developed in professional education is the capacity for collaboration and effective communication. Group collaboration is often required in vocational courses for several significant projects. Students' failure to cooperate, communicate, and divide their work reasonably inevitably impacts the group's job execution, which is detrimental to creating a peaceful and productive learning environment.

3. Storage off-site

Most instruction for vocational skills courses is conducted in a computer room, which serves as a public computer teaching

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space. However, students must continuously save data on their home and school computers. High resolution is often needed for 2D, 3D, and other works, and can only be sent using removable media such as USB flash drives. Issues with virus control and administration of computer rooms are possible. This form of remote file processing has substantial instructional implications as it necessitates the constant functioning of many computers and storage devices. Description of Function: It offers a robust and reliable storage function. Anyone with an internet connection can view any document, video, or other resource uploaded or shared by other users.

A wealth of information and resources is available to art students through interactive e-learning. Simultaneously, students may enhance their design skills and creativity via hands-on artwork discovery and production made possible by interactive e-learning's virtual reality experiences and simulation tools. Online art classes that use interactive learning strategies encourage student engagement and collaboration. Learning art is more than just teaching facts; it also allows children to practice and express themselves. By facilitating online conversation, collaborative learning, and presentation of student work, interactive e-learning allows students to connect with their instructors and other students [32]. To help students develop their design skills over time, an interactive learning environment enables them to share their work, ask questions, and receive feedback. Students' communication, teamwork, and inventive thinking skills as well as their overall quality and professional competitiveness are enhanced via this cooperative learning style. Students' unique interests and learning styles may inform the development of interactive e-learning programs that cater to their specific requirements. Students can enhance the relevance and impact of their learning by selecting learning materials and techniques based on their learning progress and objectives.

5 | A Cloud-Based Massive Data Storage Module Design

5.1 | Building the Big Data Storage Module

The quantity of data produced has skyrocketed owing to the advent of big data and rapid advancements in Internet technology. However, data sizes and types vary because consumers and data providers have different requirements. Therefore, meeting the needs of processing large-scale data analysis, for example, by making data access faster, has become crucial. The current data scale is too large for traditional data storage solutions. However, all three of these approaches have their own benefits and drawbacks, so they are not ideal for storing complicated data. One solution to these problems is a new method for storing data using cloud computing and big data. Scalable and effective data-storage solutions are among the many benefits of this paradigm. It streamlines data management and control using cloud computing to provide customers with on-demand access to storage resources [33]. Owing to the flexibility of the model, users can scale their cloud storage capacity according to their requirements. Owing to its lightning-fast data access and retrieval speeds, it is ideal for handling massive datasets. As shown in Figure 2.

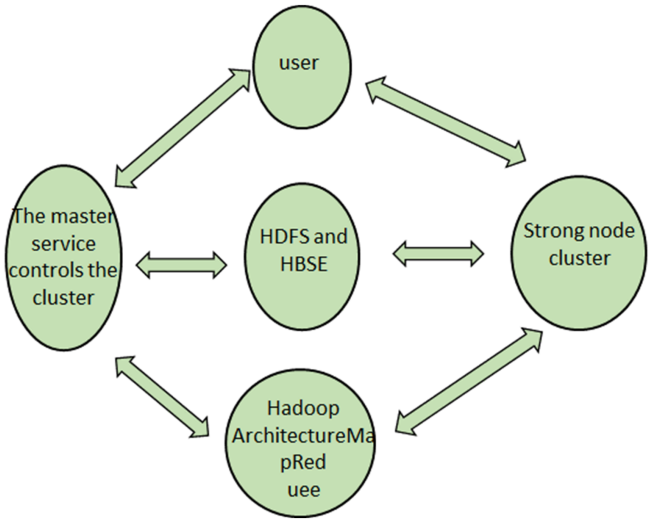


FIGURE 2 | Infrastructure for storing large amounts of cloud data.

5.2 | Enhancements and Optimizations to the Data Storage Algorithm

Using the plan from the preceding section as a foundation, the Parikh system was constructed. This approach is called the enhanced data partition storage strategy. In contrast to the first plan, we initially partition dataset d into equal-length sections r1, r2, and rN. These tasks are performed at a regional level.

The equation provides an easy way to obtain the *k*th subblock of any subblock ri as follows:

$$r_{i,k} = r_i \cdot \left(r_{i,1} \cdot r_{i,2} \cdot \cdots \cdot r_{i,k-1}\right)^{-1} \text{modp} \tag{1}$$

A different set of linear equations, (2), can be expressed by building a new linear correlation, which entails

$$\begin{aligned} a_{11}r_{1,1} + a_{12}r_{1,2} + \cdots + a_{1k}r_{1,k} &= c_{1,1} \\ a_{11}r_{2,1} + a_{12}r_{2,2} + \cdots + a_{1k}r_{2,k} &= c_{1,2} \\ &\cdots\cdots \\ a_{11}r_{n,1} + a_{12}r_{n,2} + \cdots + a_{1k}r_{n,k} &= c_{1,n} \end{aligned} \tag{2}$$

Similarly, we have

$$\begin{aligned} a_{i1}r_{1,1} + a_{i2}r_{1,2} + \cdots + a_{ik}r_{1,k} &= c_{i,1} \\ a_{i1}r_{2,1} + a_{i2}r_{2,2} + \cdots + a_{ik}r_{2,k} &= c_{i,2} \\ &\cdots\cdots \\ a_{i1}r_{n,1} + a_{i2}r_{n,2} + \cdots + a_{ik}r_{n,k} &= c_{i,n} \end{aligned} \tag{3}$$

We can determine c2, 1, C2, 2, C2, n, Ck,1, Ck, 2, and Ck, n using the same method. If c2, 1, C2, 2, C2, n, Ck, 1, Ck, 2 . . . , Ck, n, then we can write their relationship as a matrix

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kk} \end{bmatrix} \cdot \begin{bmatrix} r_{1,1} & r_{2,1} & \cdots & r_{n,1} \\ r_{1,2} & r_{2,2} & \cdots & r_{n,2} \\ \vdots & \vdots & \ddots & \vdots \\ r_{1,k} & r_{2,k} & \cdots & r_{nk} \end{bmatrix} = \begin{bmatrix} c_{1,1} & c_{1,2} & \cdots & c_{1,n} \\ c_{2,1} & c_{2,2} & \cdots & c_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{k,1} & c_{k,2} & \cdots & c_{k,n} \end{bmatrix} \tag{4}$$

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We build matrix A, matrix C, and matrix R, correspondingly, for computational ease, including

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kk} \end{bmatrix} \quad (5)$$

$$C = \begin{bmatrix} c_{1,1} & c_{1,2} & \cdots & c_{1,n} \\ c_{2,1} & c_{2,2} & \cdots & c_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{k,1} & c_{k,2} & \cdots & c_{k,n} \end{bmatrix} \quad (6)$$

$$R = \begin{bmatrix} r_{1,1} & r_{2,1} & \cdots & r_{n,1} \\ r_{1,2} & r_{2,2} & \cdots & r_{n,2} \\ \vdots & \vdots & \ddots & \vdots \\ r_{1,k} & r_{2,k} & \cdots & r_{nk} \end{bmatrix} \quad (7)$$

$$R = A^{-1}C'A^{-1} \quad (8)$$

6 | Clouds Computing-Based Large Data Security Module Design

6.1 | Development of a Large-Scale Data Security Module

Safeguarding data security and avoiding privacy infringements are essential concerns in the realm of networking during the considerable data age. This study examined data security in the context of cloud computing. The homogenous nature of cloud computing enhances its capability for security audits, testing, and assessments. The security architecture of cloud computing primarily comprises cloud customers, organizations, and third parties according to the service quality guarantees offered by prominent cloud service providers (as shown in Figure 3).

Cloud-based solutions must have strong security protocols for interactive e-learning, especially in niche areas such as art education. Protecting the cloud client against malware, trojans, and spyware begins with implementing firewalls, updates, and antivirus software. However, the overall security is significantly improved when sophisticated cloud supercomputing capabilities are used. This two-pronged strategy guarantees the ability to

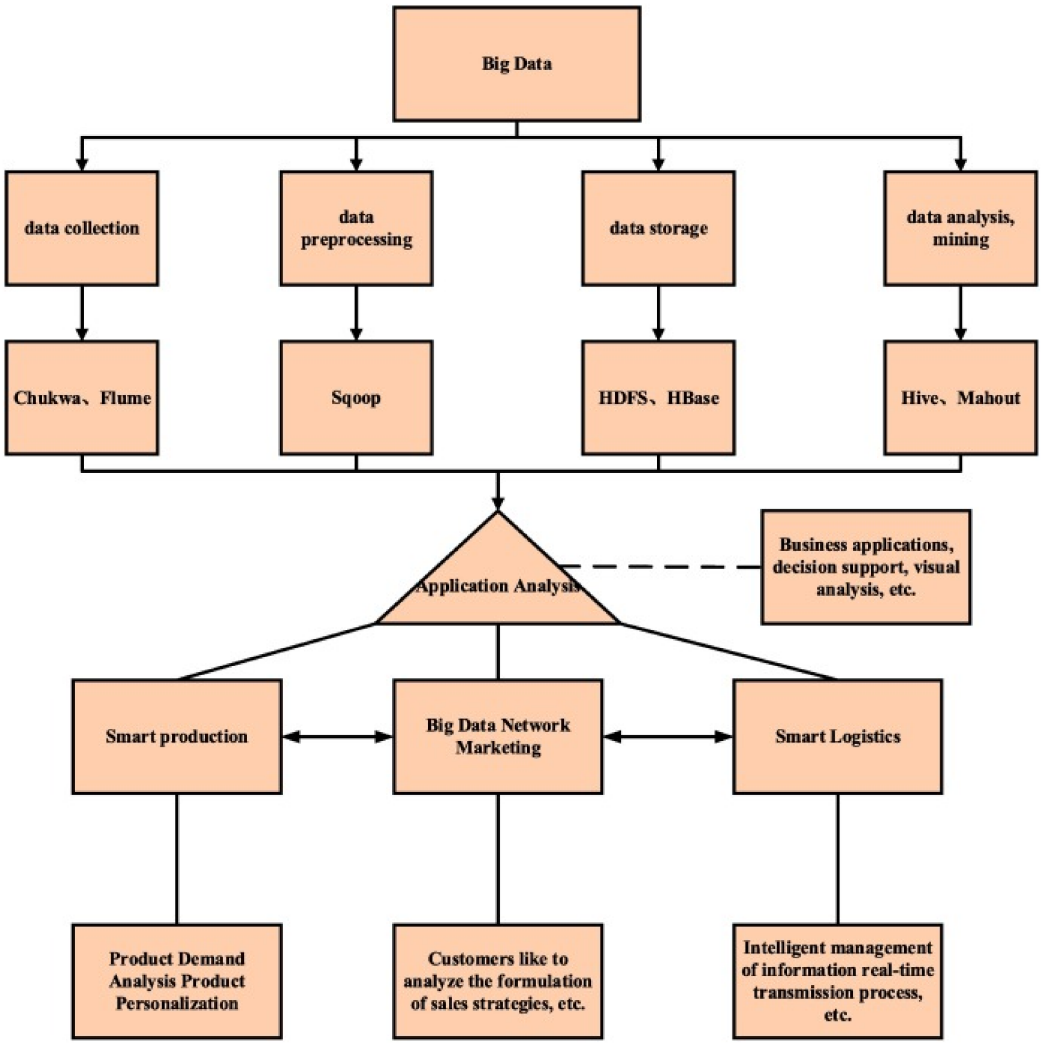


FIGURE 3 | Cloud computing security architecture.

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constantly monitor and react to new hazards and to prevent recognized threats. The capacity of cloud systems to identify unusual data flows in real time is a significant step forward in preventive security. Users can reduce the reaction time to possible attacks by sending suspicious data to a centralized background center for examination. The system prioritizes quick response and risk mitigation via the instant development of early alerts and notifications of findings to the security gateway. By streamlining the process, we can protect the system's function and integrity by documenting and managing security breaches before they escalate.

These safety precautions are essential in art education, as online resources are rapidly replacing traditional classrooms as the primary means of instruction and expression. It is necessary to safeguard personal information and creative works of art instructors and students against breaches and exploitation. By incorporating a cloud-based security architecture, the dynamic e-learning platform transforms into a stronghold for data protection, while simultaneously serving as a center for collaboration and creativity [34]. This promotes the use of digital technologies and media in educational settings by ensuring that participants can participate freely in learning and creative activities without worrying about data misuse. These security measures benefit both individual users and educational institutions. A safe and reliable learning environment is the foundation for universities to successfully create and deploy modern interactive e-learning platforms. The impact of digital media and big-data platforms on art education cannot be overstated. These innovations have revolutionized the accessibility of creative and instructional tools for educators. With the assurance that their data and intellectual property are well protected by secure systems, institutions are more likely to invest in such technologies.

Incorporating these security precautions also fits well with the bigger picture of encouraging originality and imagination in art classrooms. We can entirely free educators and students to embrace digital platform opportunities by addressing security issues. Big data platforms can propel art education when combined with instructional materials and interactive technologies. Owing to the secure and smooth access of these tools, students and teachers may benefit from uninterrupted opportunities to explore, contribute, and invent. Finally, the capabilities of big data platforms and cloud-based security measures have greatly improved the state of interactive e-learning, especially in art education [35]. These precautions can eliminate mental and technological obstacles to safe data storage and processing, thereby creating an ideal setting for digital education and innovation. In addition to safeguarding consumers and institutions, this development opens the door for art education to evolve and innovate in the digital era.

6.2 | Analysis of System Safety

In the ASA system, if a substantial data file m and its associated signature are available, the Third-Party Auditor (TPA) may accurately verify the integrity of the data block inside the large data file. Equation (9) determines the accuracy of the scheme based on the mapping properties of the bilinear pairings. In this equation, **g** represents the generator of the cryptographic system, **x** is a secret key or security parameter, and **ρ (ro)** denotes the probability measure related to system security. These components play a crucial role in ensuring the integrity and encryption strength of the data protection mechanism.

$$e(\sigma, g)^2 = \theta \cdot e\left(\prod_{i=s_1}^{s_c} h(m_i)^{v_i}, X\right) \tag{9}$$

Having established this, we demonstrated that the equation is accurate.

The right-hand side of the equation is

$$\begin{aligned} RHS &= \theta \cdot e\left(\prod_{i=s_1}^{s_c} h(m_i)^{v_i}, X\right) = \prod_{j=1}^{s} e(u_j, X)^{\mu_j} \cdot e\left(\prod_{i=s_1}^{s_e} h(m_i)^{v_i}, X\right) \\ &= \prod_{j=1}^{s} e(u_j, X)^{\sum^{v_1} v_n m_v} \cdot e\left(\prod_{i=s_1}^{s_c} h(m_i)^{v_i}, X\right) \end{aligned} \tag{10}$$

The left side of the equation is

$$\begin{aligned} LHS &= e(\sigma, g) = e\left(\prod_{i=s_1}^{s_e} \sigma_i^{v_i}, g\right) = e\left(\prod_{i=s_1}^{s_s} \left(h(m_i) \prod_{j=1}^{s} u_j^{m_i}\right)^{w_i}, g\right) \\ &= e\left(\prod_{i=\varepsilon_1}^{s_e} h(m_i)^{v_i} \cdot \prod_{j=1}^{s} \prod_{i=s_1}^{s_c} u_j^{m_v v_1}, g^x \right. \\ &= \left. e\left(\prod_{i=s_1}^{s_c} h(m_i)^{v_i} \cdot \prod_{j=1}^{s} u_j^{\sum_{i=1}^{t} v_i m_y} X\right)\right) \end{aligned} \tag{11}$$

If the conjecture regarding the CDH is correct, then the ASA method makes it computationally difficult for an attacker to generate the signature.

Calculation

$$\begin{aligned} h(m_i) \cdot \prod_{j=1}^{s} u_j^{m_v} &= \frac{g^{\gamma_i}}{\prod_{j=1}^{s} \left(g^{am_j} h^{b_j m_v}\right)} \\ \cdot \prod_{j=1}^{s} u_j^{m_y} &= \frac{g^{\gamma_i}}{\prod_{j=1}^{s} \left(g^{a_j m_v} h^{b_j m_v}\right)} \cdot \prod_{j=1}^{s} \left(g^{a_j} h^{b_j}\right)^{m_v} = g_i^{r_i} \end{aligned} \tag{12}$$

In the first step of the security model's query procedure, the attacker may determine the label of the data block MI. As a response message, the CSP inserts P = {,} for every challenge message provided by Challenger C. To satisfy Equation (13), the answer message must pass through the audit process.

$$e(\sigma, g) = \theta \cdot e\left(\prod_{i=t_1}^{s_c} h(m_i)^{v_i}, X\right) \tag{13}$$

7 | Application of Big Data Technology in Art Teaching

7.1 | Analysis of the Current Situation of Art Teaching in Colleges and Universities

Figure 4 illustrates that the persistently low employment rate of art students has led Chinese educators to recognize many deficiencies in talent cultivation within the purely academic framework of art education. Consequently, this emphasizes the need to

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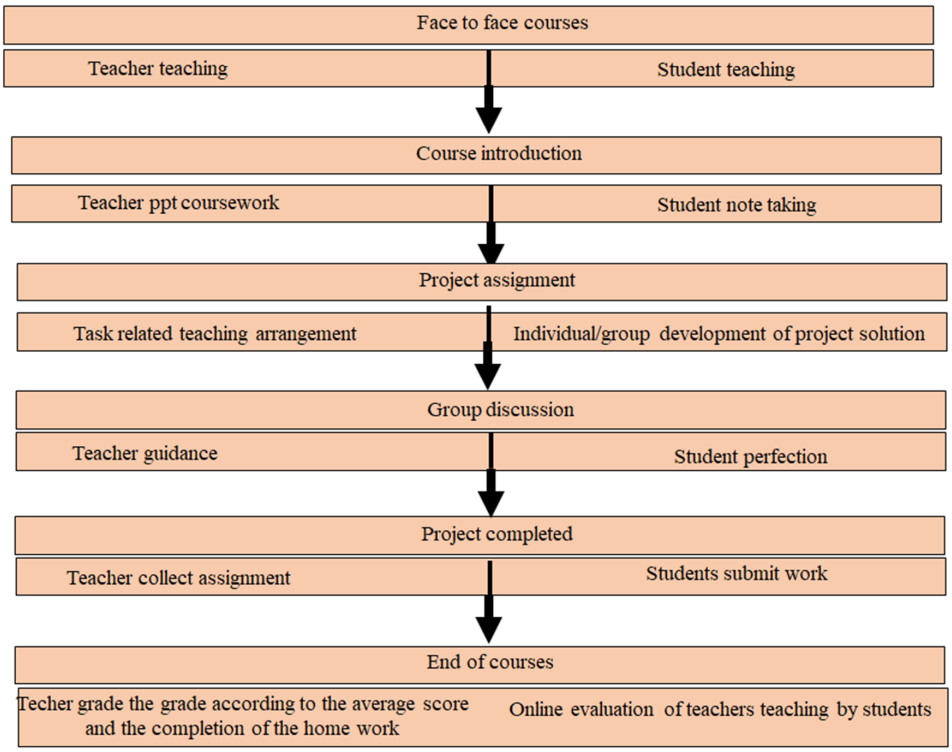


FIGURE 4 | Instructional methodology for art design major courses at domestic higher education institutions.

TABLE 1 | Application of big data and cloud computing technologies in education and professional environments.

Technology category	Students	Teachers	Practitioners
Machine learning (data mining, artificial neural networks)	12	12	12
Natural language understanding (AI translation, question answering system)	32	42	30
Computer vision (image understanding, 3D vision, dynamic vision)	27	27	37
Biometric recognition (face recognition, iris recognition, palm vein recognition)	29	20	27
Virtual reality/augmented reality	6	6	6

develop practical professional skills for art students. Numerous universities actively promote student engagement in diverse professional activities including design contests and entrepreneurial ventures. Nevertheless, the predominant approach to art education remains rooted in conventional, in-person classes. The instructor imparts knowledge in the classroom while the pupils listen attentively. The instructor assigns pertinent subjects and devises strategies for pupils based on these themes. Throughout the course, students consistently present their work under the supervision of instructors. Students may assess instructors' pedagogy and other circumstances within the after-school teaching assessment system after submitting pertinent assignments.

7.2 | Assessment of Cloud Computing and Big Data Technology Applications

Cloud computing and big data technologies are most readily applicable to art education in colleges and universities, as shown in Table 1.

Table 2 shows the distribution of responses from students, teachers, and practitioners on various cloud computing and big data technologies used in educational and professional environments.

Table 3 presents the countrywide study on colleges' and universities' attitudes toward the use of cloud computing and big data technologies in supporting art design education.

Students were the largest group of respondents; nevertheless, the proportions of students, instructors, and practitioners were roughly similar, indicating that most participants were enthusiastic about cloud computing and exhibited a significant receptivity to its use.

The findings of this study align with prior research indicating that data-driven instructional strategies improve learning outcomes. For instance, institutions that have implemented big-data-driven learning analytics have reported increased student retention and performance. Our survey results indicate that 78% of instructors observed improved engagement through personalized feedback

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TABLE 2 | Technical questionnaire on cloud computing and big data technologies used in educational and professional environments.

Technology category	Students	Teachers	Practitioners
Machine learning (data mining, artificial neural networks)	12	22	27
Natural language understanding (AI translation, question answering system)	47	15	15
Computer vision (image understanding, 3D vision, dynamic vision)	27	27	47
Biometric recognition (face recognition, iris recognition, palm vein recognition)	6	22	18
Virtual reality/augmented reality	12	16	6

TABLE 3 | Survey on the use of cloud computing and big data technologies in domestic colleges to enhance art design education.

Statement	Students	Teachers	Practitioners
Cloud computing and big data technology cannot yet be combined with art and design education	12	2	2
Cloud computing and big data technology assist in completing part of art and design education	26	30	30
Cloud computing and big data technology will be fully applied to art and design education	57	47	55
Keep observing the combination of cloud computing and big data technology with art and design education	12	30	20

mechanisms, reinforcing the practical benefits of big data applications. Additionally, qualitative responses from the faculty suggest that real-time data analysis allows them to tailor instructional strategies, thereby enhancing both teaching effectiveness and student satisfaction. These insights provide empirical justification for integrating Big Data technologies into higher education, demonstrating their feasibility for practical implementation by educators and policymakers.

7.3 | Strategic Application of Big Data Technology in Art Education

In the new paradigm, grassroots training management is more important. We should implement a management system characterized by a low center of gravity based on actual requirements, empowering grassroots units to facilitate coordination among various management components, thereby rendering the system decentralized and efficient. Conversely, the novel open training model necessitates self-correction during the training process, which requires prompt responses from grassroots units.

To establish the informatization talent training model, it is essential to devise a meticulous management framework, enhance and digitize resources across all facets in alignment with contemporary educational philosophies, systematically refine behaviors, and judiciously structure the academic and instructional processes, thereby ensuring that educational resources are adequately equipped to effectively nurture talent within the informatization context. Second, we must enhance the existing school information operational mechanism to establish an effective organizational structure and facilitate coordination of its operations.

The amalgamation of physical resources and the synthesis and application of diverse information resources provides a formidable impetus for the functionality of the new training paradigm. Historically, in the informatization development of Chinese colleges and universities, there has been a predominant focus on hardware rather than software, application infrastructure, and efficiency of service quality control. Constructing hardware serves as a "means" rather than a "purpose"; thus, we must not indiscriminately use modern technology but integrate it with practical requirements. Change via informatization cannot be achieved simply by using information technology. The data indicate that the optimal ratio of software-to-hardware investment in the informatization process should be 7:3; however, China's current ratio is 1:9. Consequently, we must focus on optimizing fundamental hardware to enhance its value, thereby improving system coordination and fostering comprehensive growth.

Developing application systems, advancing instructional applications, and training and integrating novel educational technologies must all be grounded in service-oriented teaching and practical support, enabling educators and learners to effectively assimilate into their professional environments. Enhancing teachers' quality is essential for the effective functioning of the art education system. Research findings from the United States indicate that the primary determinants influencing children's academic achievement and constraining the quality of education are the instructors themselves, far outweighing other variables such as the diverse material resources of schools. During information transformation in training mode, educators must enhance their capacity to use knowledge. Information-based education is not the only mechanical transition from conventional face-to-face instruction to online platforms. Educators must improve their comprehension of the significance of knowledge and its application while transforming their engagement with students. Numerous studies have examined the standards of teachers' abilities both domestically and internationally. As informatization advances, the educational technology proficiency of Chinese instructors must evolve into their capability to use information

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technology, necessitating a corresponding update in the criteria of these competencies. In the contemporary educational landscape, educators must possess the following competencies: choose suitable digital tools based on requirements and use technology to obtain, recognize, and manage the educational environment and apparatus; employ informational resources to comprehend the academic framework; engage with students, families, and social networks; and enhance learning efficacy.

Schools should systematically enhance instructors' proficiency in information using several methods to address deficiencies in this area. Consequently, educational institutions and educators can fully leverage the resources derived from the new training model, enhance instructors' professional development, and augment their capacity for information utilization. Establishing a contemporary team of educators is essential for effectively nurturing design skills that align with current advancements [27].

8 | Conclusion

How college instructors' education has changed dramatically due to the combination of big data and mobile Internet technology. These developments have changed conventional teaching approaches, improved opportunities for professional development, and created new options for individualized and flexible instruction. Big data have a wide range of effects on classroom growth, allowing teachers to forecast learning outcomes, evaluate student performance, and improve their methods. These features are enhanced by mobile Internet technologies, which provide real-time data availability and enable smooth teacher-student contact. These technologies work together to offer a vibrant educational environment that encourages creativity, diversity, and efficiency in higher education.

The capacity of big data to provide practical insights into student behavior and learning patterns is one of its most important contributions to improving teaching. Teachers may create focused interventions to close knowledge gaps by identifying areas where students struggle through the analysis of large volumes of data. Additionally, big data facilitates evidence-based curriculum design decision-making, enabling educators to modify their approaches to accommodate students' changing requirements. This data-driven strategy raises the overall standard of education by promoting better learning outcomes and increasing students' happiness. Mobile Internet technologies are essential for ensuring that data insights are readily available, enabling teachers to act quickly and decisively.

Professional development is another crucial area significantly influenced by big data and mobile Internet technologies. College instructors may learn new skills, exchange best practices, and work with colleagues worldwide using online training platforms and virtual communities of practice. Big data analytics can pinpoint specific areas that need improvement, making it possible to provide individualized professional development programs that cater to each person's requirements. This method fosters a culture of lifelong learning among educators and improves their teaching ability. Furthermore, mobile Internet technologies support ongoing education by offering accessible and adaptable opportunities for skill development, guaranteeing that educators stay abreast of the most recent pedagogical developments. From a practical standpoint, universities should institutionalize big data dashboards and analytics-based professional development systems to enhance evidence-driven pedagogy. Establishing continuous digital-literacy programs for faculty members can bridge gaps in data interpretation and application. Policymakers should formulate national frameworks for ethical data governance to ensure the secure, transparent, and equitable implementation of educational technologies across regions. Investment in cloud infrastructure and open-access digital platforms will further democratize the benefits of big data innovations for both educators and students.

Notwithstanding these advantages, there are several drawbacks to using big data and mobile Internet technologies for educational development. As many instructors need more ability to evaluate and comprehend extensive statistics properly, data literacy remains a significant obstacle. Targeted training programs and professional development activities are necessary to address this problem and provide instructors with the skills required to use big data properly. Another important issue is privacy and security, primarily because of the growing dependence on digital platforms for data processing and storage. To secure sensitive data and foster stakeholder confidence, implementing strong data protection policies and ethical standards is essential.

Additionally, the digital divide is a significant barrier to the fair integration of big data and mobile Internet technologies in education. By restricting the advantages of these developments to privileged institutions and areas, disparities in access to infrastructure and technology may worsen existing inequities. To close this gap, legislators, academic institutions, and tech companies must work together to guarantee that all teachers have the tools and assistance required to incorporate big data into lesson plans.

This study provides empirical insights into how big data-driven educational technologies enhance instructional quality. However, this study has some limitations. First, it focuses on higher-education institutions in China, which may limit the generalisability of the findings to other educational systems. Second, data privacy and literacy concerns remain significant barriers that require further policy-driven intervention. Future research should explore the scalability of big data tools in resource-constrained settings and assess their long-term impact on student performance. Future research should pursue multi-country comparative studies and longitudinal tracking to assess the sustained pedagogical and ethical effects of big data adoption in higher education. Expanding the sample beyond Chinese institutions and integrating student, administrative, and policy perspectives will enhance generalisability. Future work could employ experimental or quasi-experimental designs to quantify the causal impact of big-data analytics on instructional outcomes and teacher professional growth.

In summary, big data and mobile Internet technologies have transformed college instructors' learning experiences and provided creative solutions to persistent problems in higher education. These technologies offer instructors the ability to provide high-quality instruction adapted to students' various

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requirements by facilitating data-driven decision-making, individualized learning, and ongoing professional development. However, to fully benefit from these developments, issues related to accessibility, privacy, and data literacy must be resolved. Educational institutions may provide a welcoming and encouraging atmosphere that encourages the efficient use of big data and mobile Internet technologies in the classroom by investing in infrastructure, training, and creating regulatory frameworks. As these technologies continue to develop, they have the potential to make education more effective, fair, and enjoyable for teachers and students.

Despite its contributions, this study has several limitations. First, it focused on higher-education institutions in China, which may limit the generalisability of the findings to other educational systems. Second, while big data has demonstrated its potential for improving instructional effectiveness, challenges remain regarding ethical considerations, digital literacy among faculty, and accessibility in resource-constrained environments. Future research should explore the long-term impact of big data-driven education on student learning outcomes and faculty development. Additionally, studies should focus on developing standardized frameworks for integrating big data analytics into diverse educational settings to ensure equity and effectiveness.

Author Contributions

Ling Yu: methodology, conceptualization, project administration, resources, writing – original draft, writing – review and editing. **Wenye Li:** methodology, conceptualization. **Ying Luo:** conceptualization, investigation, writing – original draft.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Research data are not shared.

References

1. P. Gao, "System Design of a Vocal Music Teaching Platform Based on Virtual Reality Technology," *Lecture Notes in Electrical Engineering* 1031 (2023): 587–594, https://doi.org/10.1007/978-981-99-1428-9_72.

2. M. S. Nascimento, H. O. Lima, J. C. F. Lima, A. Z. S. de Freitas, and A. P. Mendonça, "Garbage in the Public Space: Sensitization by Science Teaching Using Virtual Reality," *International Journal for Innovation Education and Research* 7, no. 12 (2019): 313–325, https://doi.org/10.31686/IJIER.VOL7.ISS12.2058.

3. I. Adeleye, J. Luiz, J. Muthuri, and K. Amaeshi, "Business Ethics in Africa: The Role of Institutional Context, Social Relevance, and Development Challenges," *Journal of Business Ethics* 161, no. 4 (2020): 717–729, https://doi.org/10.1007/s10551-019-04338-x.

4. S. Roscoe, N. Subramanian, C. J. C. Jabbour, and T. Chong, "Green Human Resource Management and the Enablers of Green Organisational Culture: Enhancing a Firm's Environmental Performance," *Business Strategy and the Environment* 28, no. 5 (2019): 737–749, https://doi.org/10.1002/bse.2277.

5. Y. Chen and H. Zheng, "The Application of HMM Algorithm-Based Music Note Feature Recognition Teaching in Universities," *Intelligent Systems with Applications* 20 (2023): 200277, https://doi.org/10.1016/j.iswa.2023.200277.

6. K. M. Chuah and M. K. Kabilan, "Teachers' Views on the Use of Chatbots to Support English Language Teaching in a Mobile Environment," *International Journal of Emerging Technologies in Learning* 16, no. 20 (2021): 223–237, https://doi.org/10.3991/ijet.v16i20.24917.

7. L. Lingyan and G. Yukun, "Visual Analysis of Undergraduate Teaching in Business Administration," *Procedia Computer Science* 199 (2022): 788–797, https://doi.org/10.1016/j.procs.2022.01.098.

8. S. Wearne, "Stakeholders in Excellence in Teaching and Learning of Project Management," *International Journal of Project Management* 26, no. 3 (2008): 326–328, https://doi.org/10.1016/j.ijproman.2008.02.002.

9. A. Setiawan, F. Agiwahyuanto, and P. Arsiwi, "A Virtual Reality Teaching Simulation for Exercise During Pregnancy," *International Journal of Emerging Technologies in Learning* 14, no. 1 (2019): 34–48, https://doi.org/10.3991/ijet.v14i01.8944.

10. W. Leal Filho, P. C. C. Ribeiro, J. Mazutti, et al., "Using Artificial Intelligence to Implement the UN Sustainable Development Goals at Higher Education Institutions," *International Journal of Sustainable Development and World Ecology* 31 (2024): 726–745, https://doi.org/10.1080/13504509.2024.2327584.

11. H. Xie, H. C. Chu, G. J. Hwang, and C. C. Wang, "Trends and Development in Technology-Enhanced Adaptive/Personalized Learning: A Systematic Review of Journal Publications From 2007 to 2017," *Computers & Education* 140 (2019): 103599, https://doi.org/10.1016/j.compedu.2019.103599.

12. J. Cabero-Almenara, C. Llorente-Cejudo, and R. Martinez-Roig, "The Use of Mixed, Augmented, and Virtual Reality in History of Art Teaching: A Case Study," *Applied System Innovation* 5, no. 3 (2022): 44, https://doi.org/10.3390/asi5030044.

13. P. Jain, M. Gyanchandani, and N. Khare, "Big Data Privacy: A Technological Perspective and Review," *Journal of Big Data* 3, no. 1 (2016): 1–25, https://doi.org/10.1186/s40537-016-0059-y.

14. A. Cuzzocrea, "Privacy and Security of Big Data: Current Challenges and Future Research Perspectives," Proceedings of the First International Workshop on Privacy and Security of Big Data (pp. 45–47) 2014, https://doi.org/10.1145/2663715.2669614.

15. B. Hamamra, Z. N. Khlaif, and A. Mayaleh, "Between Promise and Precarity: Palestinian University Educators' Experiences With AI Integration in Higher Education," *Interactive Learning Environments* 33 (2025): 1–16, https://doi.org/10.1080/10494820.2025.2526435.

16. A. Ayyoub, Z. Khlaif, B. Hamamra, et al., "Drivers of Acceptance of Generative AI Through the Lens of the Extended Unified Theory of Acceptance and Use of Technology," *Human Behavior and Emerging Technologies* 2025, no. 1 (2025): 6265087, https://doi.org/10.1002/hbe2.5087.

17. B. Hamamra and A. Qabaha, "'Can the Subaltern Speak?': COVID-19 and Decolonial Pedagogy in Palestinian Universities," *Journal for Cultural Research* 25, no. 2 (2021): 189–201, https://doi.org/10.1080/14797585.2021.1936106.

18. B. Hamamra, Z. N. Khlaif, A. Mayaleh, and A. A. Baker, "A Phenomenological Examination of Educators' Experiences With AI Integration in Palestinian Higher Education," *Cogent Education* 12, no. 1 (2025): 2526435, https://doi.org/10.1080/2331186X.2025.2526435.

19. G. Chowdhury and K. Koya, "Information Practices for Sustainability: Role of iSchools in Achieving the UN Sustainable Development Goals (SDGs)," *Journal of the Association for Information Science and Technology* 68, no. 9 (2017): 2128–2138, https://doi.org/10.1002/asi.23825.

20. S. Sakibayev, R. Sakibayev, and B. Sakibayeva, "The Educational Impact of Using Mobile Technology in a Database Course in College," *Interactive Technology and Smart Education* 16, no. 4 (2019): 363–380, https://doi.org/10.1108/itse-02-2019-0021.

21. B. D. Rodney, "Understanding the Paradigm Shift in Education in the Twenty-First Century: The Role of Technology and the Internet of

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Things," *Worldwide Hospitality and Tourism Themes* 12, no. 1 (2020): 35–47, https://doi.org/10.1108/whatt-10-2019-0066.

22. W. R. Penuel, R. Riedy, M. S. Barber, D. J. Peurach, W. A. LeBouef, and T. Clark, "Principles of Collaborative Education Research With Stakeholders: Toward Requirements for a New Research and Development Infrastructure," *Review of Educational Research* 90, no. 5 (2020): 627–674, https://doi.org/10.3102/0034654320938126.

23. N. Marangunić and A. Granić, "Technology Acceptance Model: A Literature Review From 1986 to 2013," *Universal Access in the Information Society* 14, no. 1 (2015): 81–95, https://doi.org/10.1007/s10209-014-0348-1.

24. D. Ussher-Eke, "HR and GDPR: Partnering to Protect Employee Data," *World Journal of Advanced Research and Reviews* 27, no. 2 (2025): 717–730, https://doi.org/10.30574/wjarr.2025.27.2.0583.

25. A. Kumar, "Productizing Pedagogy: Educational Television, Wicked Problems, and the Study of Social Innovation," preprint, arxiv, 2025, https://doi.org/10.48550/arXiv.2503.01987.

26. S. Wu, Y. Cao, J. Cui, et al., "A Comprehensive Exploration of Personalized Learning in Smart Education: From Student Modeling to Personalized Recommendations," preprint, arXiv, 2024, arXiv:2402.01666.

27. M. P. Arunachalam, "A Secure and Efficient Data Store and Remote Data Integrity Checking in Cloud Computing Using TPA," *International Journal of Cloud Computing* 12, no. 3 (2024): 244–259, https://doi.org/10.1504/IJCC.2024.100123.

28. A. Z. Sarnato, W. D. Sari, S. T. Rahmawati, R. Hidayat, and H. Patry, "The Evolution of e-Learning Platforms: From u-Learning to AI-Driven Adaptive Learning Systems," *Journal of Social Science Utilizing Technology* 2, no. 2 (2024): 289–300.

29. S. S. Bakare, A. O. Adeniyi, C. U. Akpuokwe, and N. E. Eneh, "Data Privacy Laws and Compliance: A Comparative Review of the EU GDPR and USA Regulations," *International Journal of Law and Policy* 5, no. 1 (2024): 88–104.

30. T. M. Kolade, N. T. Aideyan, S. M. Oyekunle, O. S. Ogungbemi, D. L. Dapo-Oyewole, and O. O. Olaniyi, "Artificial Intelligence and Information Governance: Strengthening Global Security Through Compliance Frameworks and Data Security," *SSRN Electronic Journal* 17 (2024), https://doi.org/10.2139/ssrn.5044032.

31. R. Mojtahedzadeh, S. Hasanvand, A. Mohammadi, S. Malmir, and M. Vatankhah, "Students' Experience of Interpersonal Interactions Quality in e-Learning: A Qualitative Research," *PLoS One* 19, no. 3 (2024): e0298079, https://doi.org/10.1371/journal.pone.0298079.

32. S. Guan, C. Zhang, Y. Wang, and W. Liu, "Hadoop-Based Secure Storage Solution for Big Data in Cloud Computing Environment," *Digital Communications and Networks* 10, no. 1 (2024): 227–236, https://doi.org/10.1016/j.dcan.2023.07.008.

33. J. A. Anitha, "Revolutionizing Education: The Multifaceted Role of Cloud in Modern e-Learning Environment," in *Cloud Computing for Smart Education and Collaborative Learning*, ed. M. Gupta (Chapman & Hall/CRC, 2025), 286–298, https://doi.org/10.1201/9781003456789-16.

34. A. Roziqin, "Insights Into Digital Business, Human Resource Management, and Competitiveness in the Private and Public Sectors in Indonesia," *Journal of Business Innovation* 5, no. 1 (2025): 377–395.

35. Y. He, "Graduates' Employability in the Creative Industry in China: What Competencies, Qualities, and Skills Chinese Graduates With an Undergraduate Degree in Fine Art Need for Employment in China," (Doctoral diss., University for the Creative Arts, 2024), https://ucreative.edu/theses/he-2024-fine-art-employment.

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